Fundamentals 22

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Problem 22.1

Let

$$f(x) = \begin{cases} 1 & \text{if } x = 1, \\ 2 & \text{otherwise,} \end{cases}$$

$$A = \{1, 2\}, \quad B = \{1, 3\}.$$

Then

$$f(A \cap B) = f(\{1\}) = \{1\}$$

but

$$f(A) \cap f(B) = \{1, 2\} \cap \{1, 2\} = \{1, 2\}.$$

Problem 22.2

Part (1)

Let $y \in f(f^{-1}(V))$. This means that there is some $x \in f^{-1}(V)$ such that $f(x) = y$. Since $x \in f^{-1}(V)$, $f(x) \in V$. But $y = f(x)$, so $y \in V$. So $f(f^{-1}(V)) \subseteq V$.

Part (2)

Let $x \in U$. Then $f(x) \in f(U)$. Let $y = f(x)$. Since $y \in f(U)$, all x' such that $f(x') = y$ is in $f^{-1}(f(U))$. Since $f(x) = y$, $x \in f^{-1}(f(U))$. So $U \subseteq f^{-1}(f(U))$.

Problem 22.3

First, we show that $f^{-1}(V \backslash B) = A \backslash f^{-1}(V)$. Assume $x \in A \backslash f^{-1}(V)$. Then $x \notin f^{-1}(V)$, so $f(x) \notin V$. Then, if $f(x) \notin V$, $x \notin f^{-1}(V)$. So $x \in A \backslash f^{-1}(V)$, and therefore, $f(x) \notin V$ iff $x \in A \backslash f^{-1}(V)$.

Assume $x \in f^{-1}(B \backslash V)$. Then $f(x) \in B \backslash V$ so $f(x) \notin V$. If $f(x) \notin V$, then $f(x) \in B \backslash V$ so $x \in f^{-1}(B \backslash V)$. So $f(x) \notin V$ iff $x \in f^{-1}(B \backslash V)$.

Since $f(x) \notin V$ iff $x \in A \backslash f^{-1}(V)$ and iff $x \in f^{-1}(B \backslash V)$, $A \backslash f^{-1}(V) = f^{-1}(B \backslash V)$.

Now assume $f : A \to B$ is continuous and let $V \subseteq B$ be closed. Then $B \backslash V$ is open. Let $V' = B \backslash V$; since f is continuous, $f^{-1}(V')$ is open. Therefore,

$$A \backslash f^{-1}(V') = A \backslash f^{-1}(B \backslash V) = A \backslash (A \backslash f^{-1}(V)) = f^{-1}(V)$$

is closed.

Now assume that $f^{-1}(V)$ is closed in A if V is closed in B. Let $V \subseteq B$ and $V' = B \backslash V$. We have that $f^{-1}(B \backslash V')$ is closed if $B \backslash V'$ is closed, so $A \backslash f^{-1}(B \backslash V')$ is open if V' is open. Since $A \backslash f^{-1}(B \backslash V') = f^{-1}(V')$, we have that $f^{-1}(V')$ is open if V' is open, so f is continuous.

Problem 22.4

Let $f : X \to Y$ be continuous. Fix x and a neighborhood V where $f(x) \in V$, and let $U = f^{-1}(V)$. Since V is open, U is open, so $f(U) = f(f^{-1}(V)) \subseteq V$.

Now, assume that for all x and each neighborhood V of $f(x)$ there is U such that U is open and $x \in U$ such that $f(U) \subseteq V$. Let $V \subseteq Y$ be open. For each $x_\alpha \in f^{-1}(V)$, U_α is open with $x \in U_\alpha$ and $f(U_\alpha) \subseteq V$. Let $U = \bigcup_{\alpha \in A} U_\alpha$. We claim that $U = f^{-1}(V)$.

If $x_\alpha \in f^{-1}(V)$ then $x_\alpha \in U$, so $f^{-1}(V) \subseteq U$. If $x \in U$, then $x \in U_\alpha$ for some $\alpha \in A$ and $f(U_\alpha) \subseteq V$, so for all $x \in U_\alpha$, $f(x) \in V$, so $x \in f^{-1}(V)$. So $f^{-1}(V) = U$. Since U is a union of open intervals, it is open, so $f^{-1}(V)$ is open, and f is continuous.

Problem 22.5

Let V be open. Then $f^{-1}(V)$ is open since f is continuous. Then, since g is continuous, $g^{-1}(f^{-1}(V))$ is also open.

Problem 22.7

Part (1)

1. We have that $\varnothing \in T$ and $\varnothing$ is finite, so $\mathbb{R} \backslash \varnothing = \mathbb{R} \in \mathcal{T}$.
2. Let $\{X_\alpha : \alpha \in A\}$ be a subset of $\mathcal{T}$. Then $\mathbb{R} \backslash X_\alpha$ is finite. We have that
$$\mathbb{R} \backslash \bigcup_{\alpha \in A} X_\alpha = \bigcap_{\alpha \in A} (\mathbb{R} \backslash X_\alpha),$$
which is finite, so $\bigcup_{\alpha \in A} X_\alpha \in \mathcal{T}$.
3. Let $Y, Z \in \mathcal{T}$, so $\mathbb{R} \backslash Y$ and $\mathbb{R} \backslash Z$ are finite. Then $\mathbb{R} \backslash (Y \cap Z) = (R \backslash Y) \cup (\mathbb{R} \backslash Z)$ which is finite, so $Y \cap Z \in \mathcal{T}$.

Part (2)

Let

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ -x & \text{otherwise.} \end{cases}$$

Let V be open, so that $\mathbb{R}\backslash V$ is finite. Then f has an inverse: $f^{-1}(x) = -f(x)$, so $|\mathbb{R}\backslash f^{-1}(V)| = |\mathbb{R}\backslash V|$ which is finite, so $f^{-1}(V)$ is open. This function is interesting because it's not continuous under the standard topology on $\mathbb{R}$.

Part (3)

Let $f(x) = \sin(x)$. This is discontinuous, since we can let $V = \mathbb{R}\backslash\{0\} \in \mathcal{T}$, and we have that $f^{-1}(V) = \mathbb{R}\backslash\{\pi k : k \in \mathbb{Z}\} \notin \mathcal{T}$. This function is interesting because it's continuous under the standard topology on $\mathbb{R}$.

Problem 22.8

Let $U \subseteq \mathbb{R}$ be open. We want to show that

$$U = \bigcup_{\alpha \in A} U_\alpha$$

for some countable set A and open intervals U_α.

For all $x \in U$, define

$$a = \inf\{y \in \mathbb{R} : y < x \text{ and } [x, y] \subseteq U\},$$

$$b = \sup\{y \in \mathbb{R} : y > x \text{ and } [x, y] \subseteq U\},$$

where possibly $a = -\infty$ or $b = \infty$.

Then the set $U_x = (a, b)$ is the largest open interval with $x \in U_x \subseteq U$. Let $\{U_\alpha : \alpha \in A\}$ contain every unique U_x for all $x \in U$. Then for any $x \in U$ there is $\alpha \in A$ such that $x \in U_\alpha$. So $U = \bigcup_{\alpha \in A} U_\alpha$. Also, all elements of U_α are disjoint by construction.

Every nonempty open interval contains a rational number, so we can define a function $f : \{U_\alpha : \alpha \in A\} \to \mathbb{Q}$ by picking for each U_α a rational $q \in U_\alpha$. Because the U_α are disjoint, the function is injective: if $U_\alpha \neq U_\beta$ then $q_\alpha \neq q_\beta$. Since $\mathbb{Q}$ is countable and f is an injection, $\{U_\alpha : \alpha \in A\}$ is either finite or countable.